

WORLD WAR II LONG-RANGE HEAVY BOMBER

BOEING B-17 FLYING FORTRESS

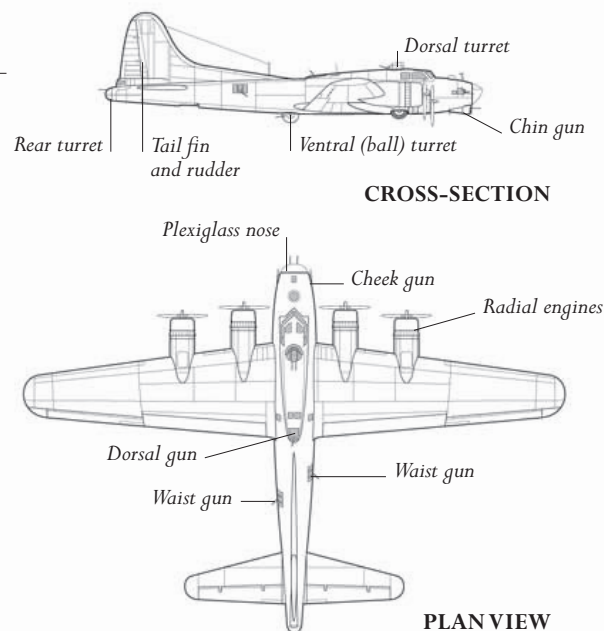
Conceived in the 1930s, the B-17 epitomized a new type of bomber aircraft, able to carry heavy loads over long distances. Supplied to Great Britain by the US as part of the Lend Lease deal, it became one of the mainstays of the Allied bombing offensive in Europe.

The B-17C was the first version to go to war, in July 1941, but only with the arrival of the B-17E two months later did the aircraft begin to make its mark. Its reputation was assured when the definitive B-17G was introduced in October 1943.

For short-range missions, a B-17G could carry a 3,600kg (8,000lb) bomb load, but that was reduced to 2,700kg (6,000lb)

when the aircraft was fuelled for 3,200km (2,000 miles). The top speed was 462kph (287mph), although in mass formation the planes did not exceed 290kph (180mph).

Flying by day made B-17s vulnerable, in spite of their heavy armament, but losses fell after long-range fighters were developed to escort them. In all, 12,671 B-17s were built, and around 37 per cent of these were lost.



▲ BOMBER PROFILE

The B-17 had a wingspan of 31.62m (103¾ft) and measured 22.66m (74⅓ft) from nose to tail. The huge tail fin gave the aircraft a distinctive profile.

THE STRUCTURE



▲ ROBUST UNDERCARRIAGE

To support a loaded weight of almost 25.5 tonnes (25 tons), the undercarriage struts had to be robust, with massive hydraulic shock-absorbers.

▲ ALUMINIUM SKIN

The wings and fuselage were clad with light-gauge aluminium, riveted to the frames beneath.



▲ WING ROOT

Construction was not sophisticated; no effort was made to improve the airflow by counter-sinking the fastenings.

► RADIAL ENGINE

The B-17's four 1,200hp Wright radial engines were equipped with turbo-superchargers to improve high-altitude performance.

